Friedrich-Alexander-Universität
Technische Fakultät

What's next in medical AI?

From code to clinic – at startup speed

Andreas Maier

Friedrich-Alexander-Universität Erlangen-Nürnberg 12 June 2026 • DLDxHealth, Munich

Outline

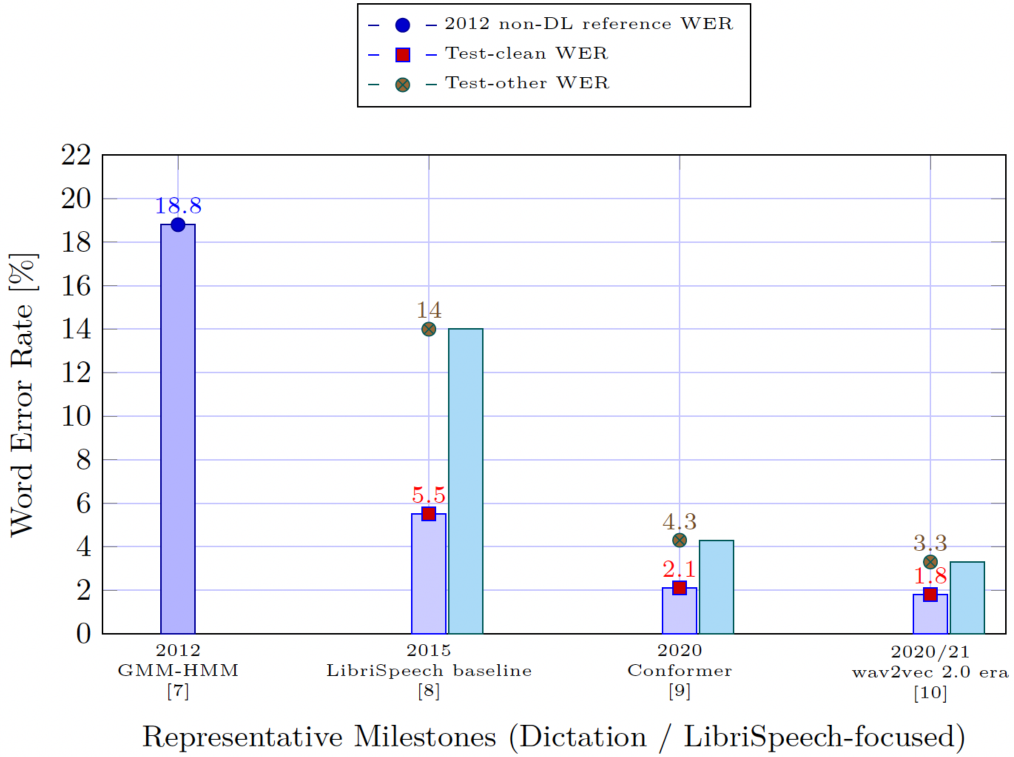
- What is deep learning?
- What is agentic AI?
- What is coming next?

What is deep learning?

Deep learning

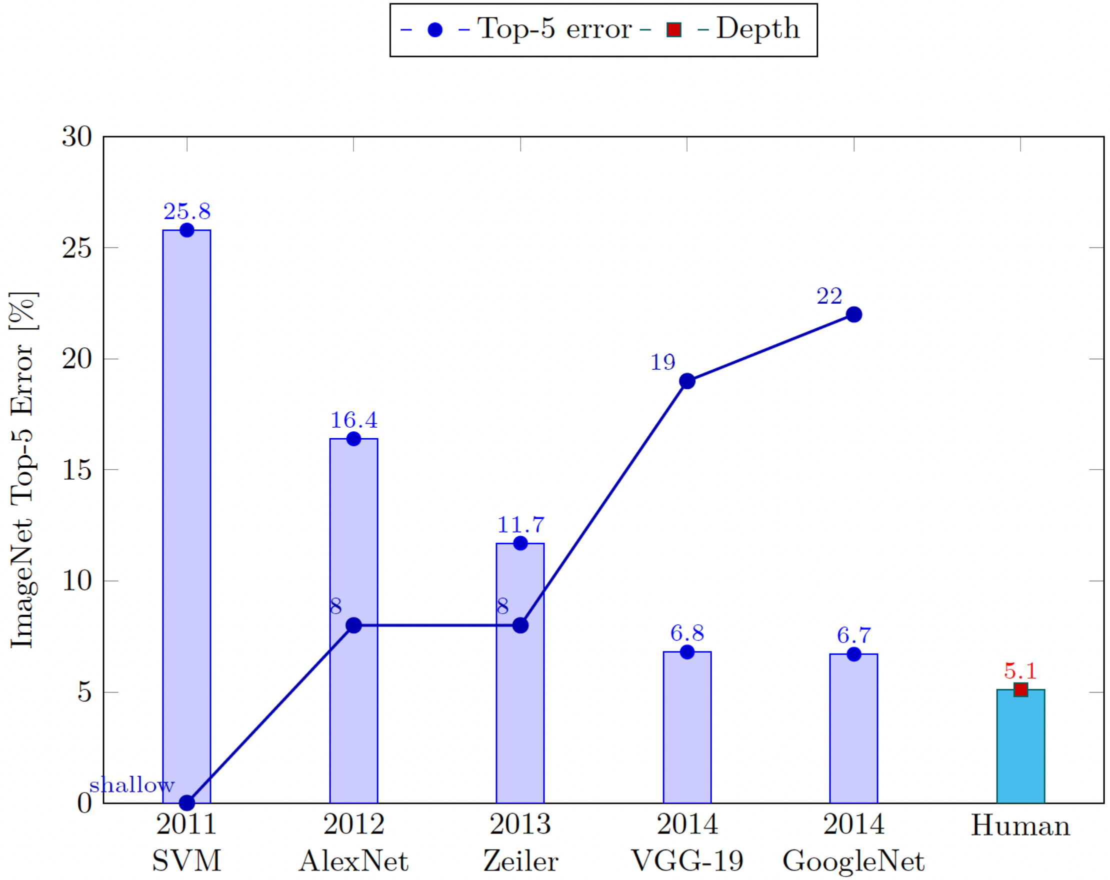
Speech

- Revolutionising pattern recognition and AI for more than 10 years.



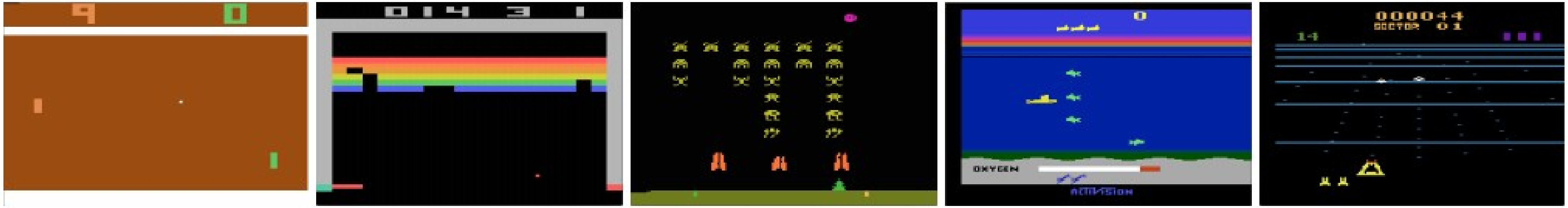
Deep learning

Image classification



Deep learning

Learning to play

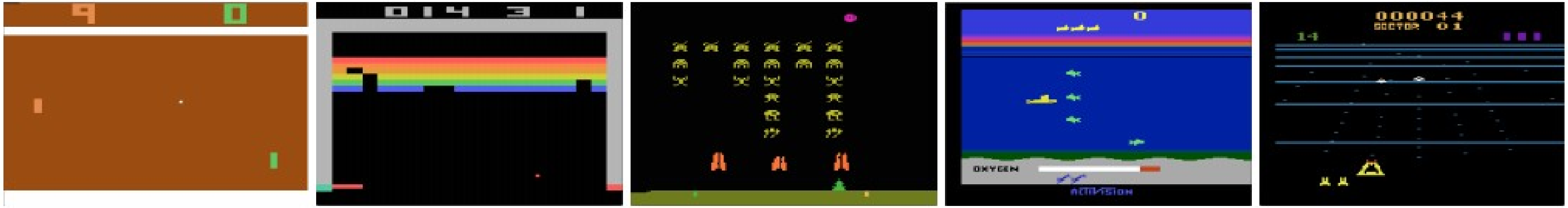


[Mnih 2013]

- Deep Q-Networks learn Atari games from raw pixels alone.
- One agent, many games – super-human on several titles.

Deep learning

Learning to play



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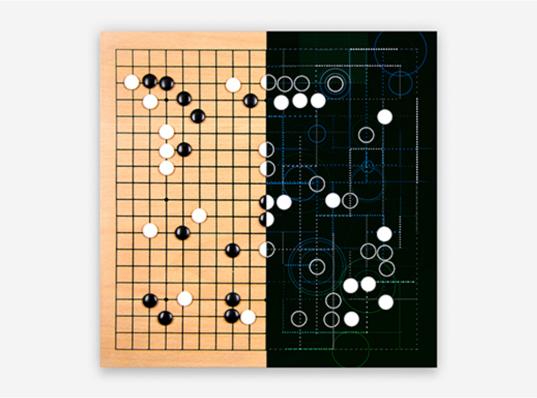


AlphaGo (DeepMind, 2016)

- Self-play + Monte-Carlo tree search.
- Defeats world champion Lee Sedol 4–1.

Generative AI

Source image

Generative AI

Image description

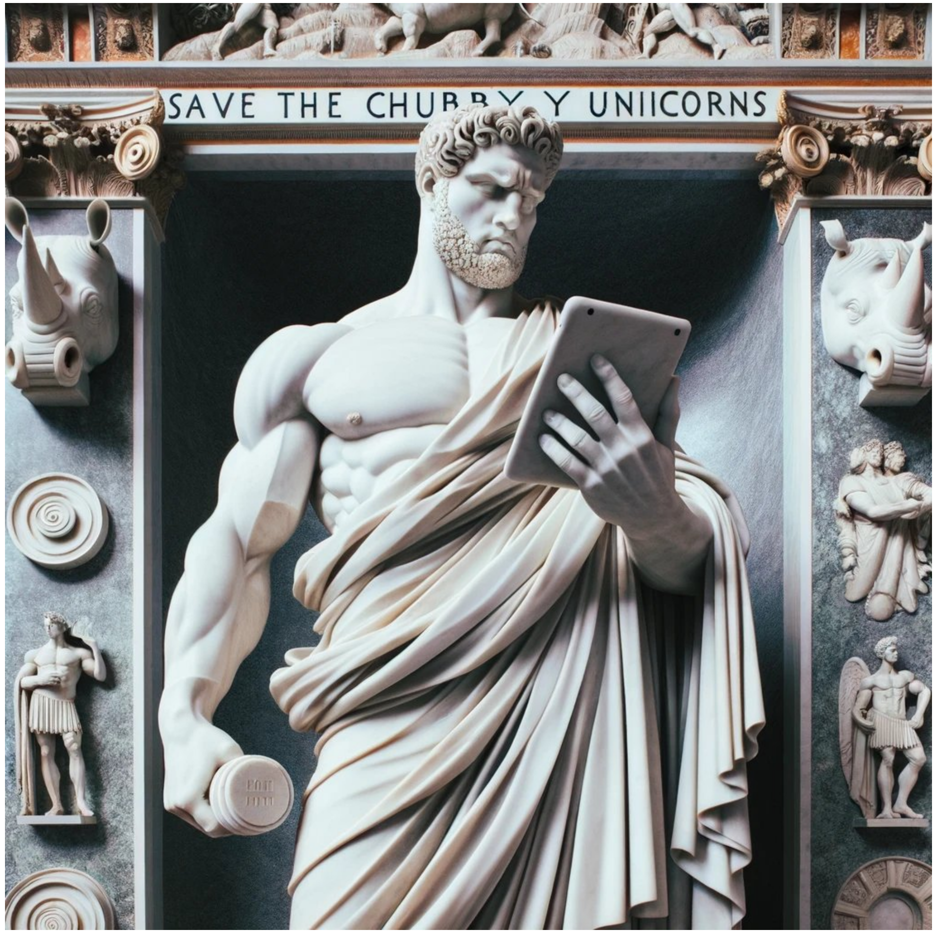
"An athletically built man with a serious expression stands in a well-lit room with a wooden ceiling. He is wearing a blue t-shirt with the phrase 'SAVE THE CHUBBY UNICORNS' and an image of a rhinoceros. The man is taking a selfie with a modern smartphone, capturing his reflection in a mirror. In the background, one can see gym equipment including dumbbells and a weight rack. There is also a hanging chandelier with a unique design, illuminating the room."

Generative AI

Generated image

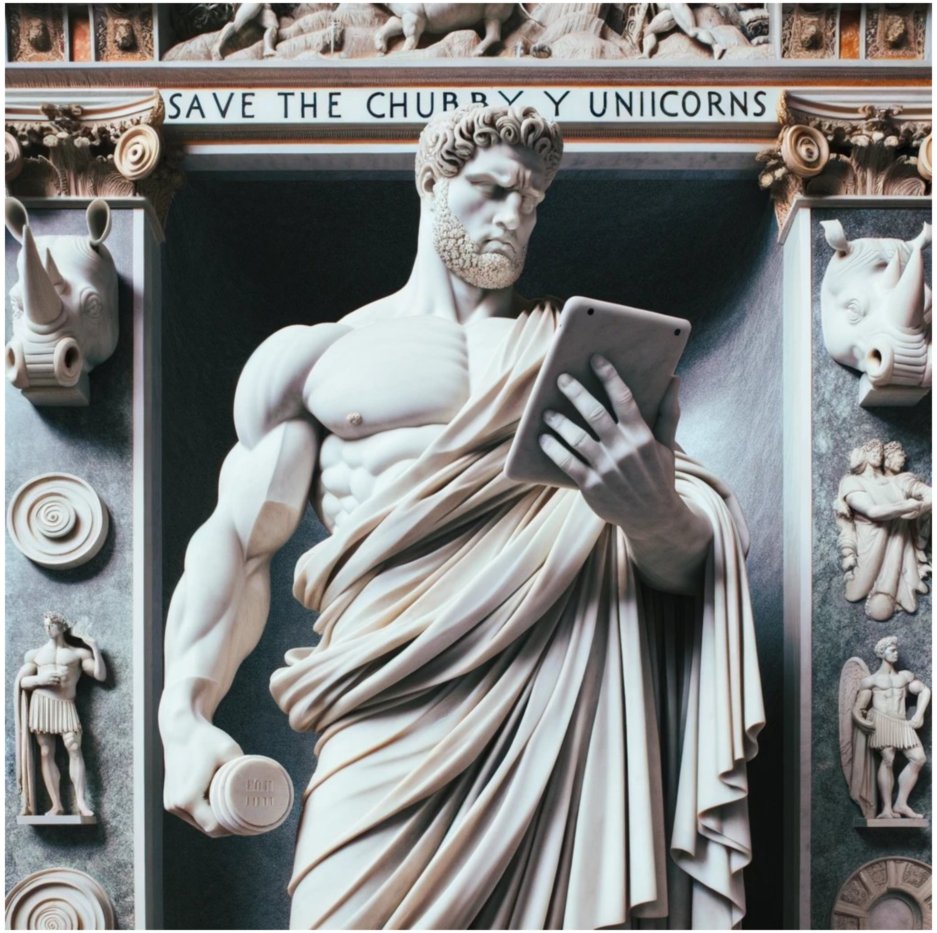
Generative AI

Style variations



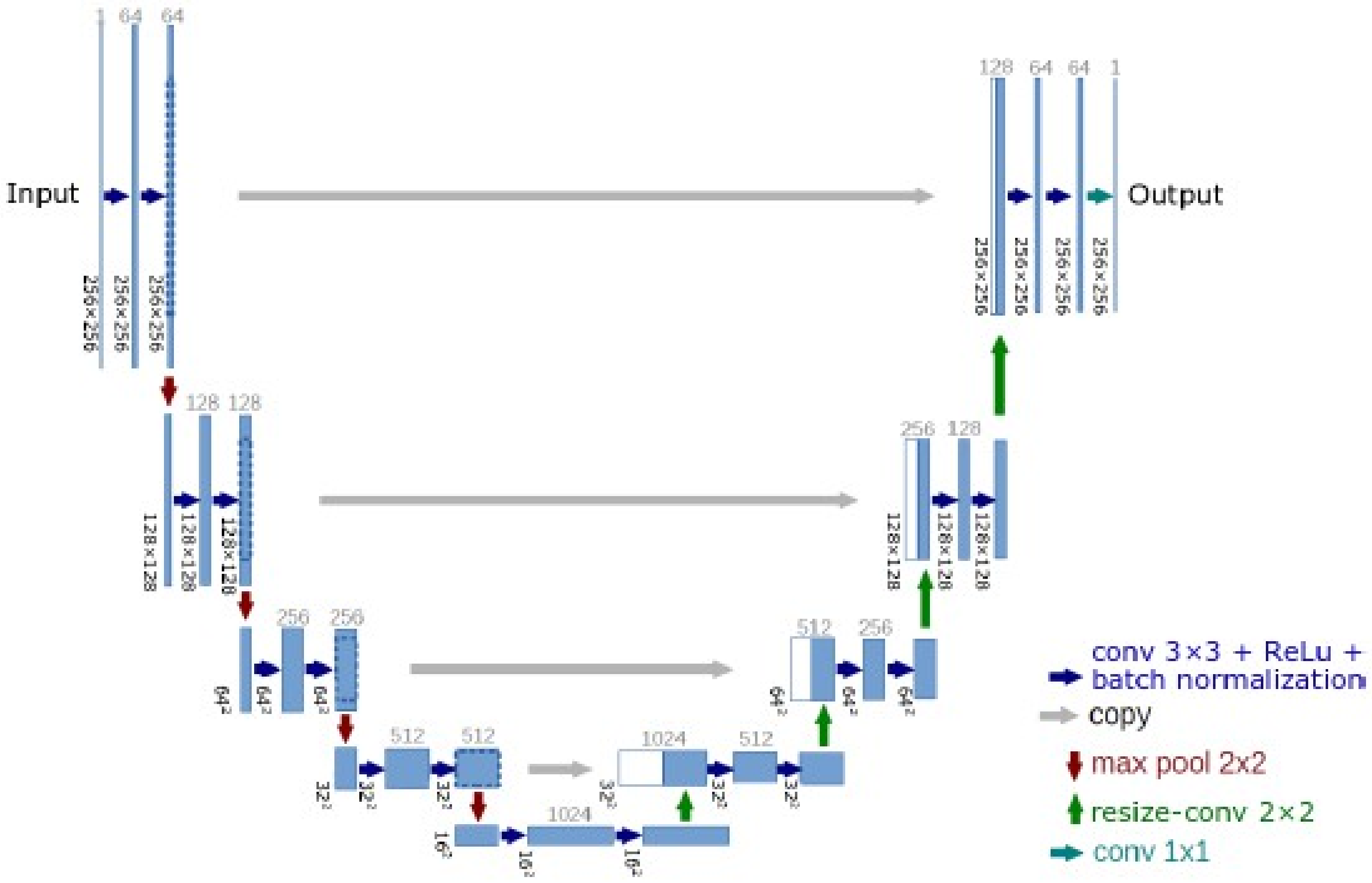
Generative AI

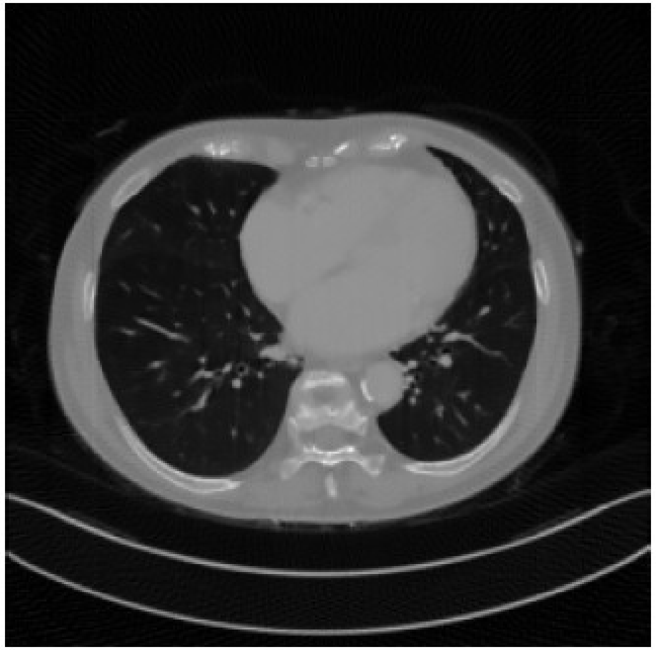
Style variations



Deep learning in medicine

From Segmentation to Reconstruction





Deep learning in medicine

Known operators

$$x \rightarrow \boxed{\begin{array}{c}\text{Trainable}\\ \hat{f}_1\end{array}} \rightarrow \boxed{\begin{array}{c}\text{Known op.}\\ f_2\end{array}} \rightarrow \boxed{\begin{array}{c}\text{Trainable}\\ \hat{f}_3\end{array}} \rightarrow \boxed{\begin{array}{c}\text{Known op.}\\ f_4\end{array}} \rightarrow \hat{f}(x)$$

Mix *trainable* layers with deterministic operators (Fourier transform, gradient descent, projection, . . .).

Deep learning in medicine

Known operators

$x \rightarrow$ Trainable $\hat{f}_1$ $\rightarrow$ Known op. f_2 $\rightarrow$ Trainable $\hat{f}_3$ $\rightarrow$ Known op. f_4 $\rightarrow \hat{f}(x)$

Mix *trainable* layers with deterministic operators (Fourier transform, gradient descent, projection, ...).

Theorem – known operators reduce the error bound

For $f = f_n \circ \cdots \circ f_1$ approximated by $\hat{f} = \hat{f}_n \circ \cdots \circ \hat{f}_1$ with per-layer errors ε_i and Lipschitz constants $\hat{L}_j$,

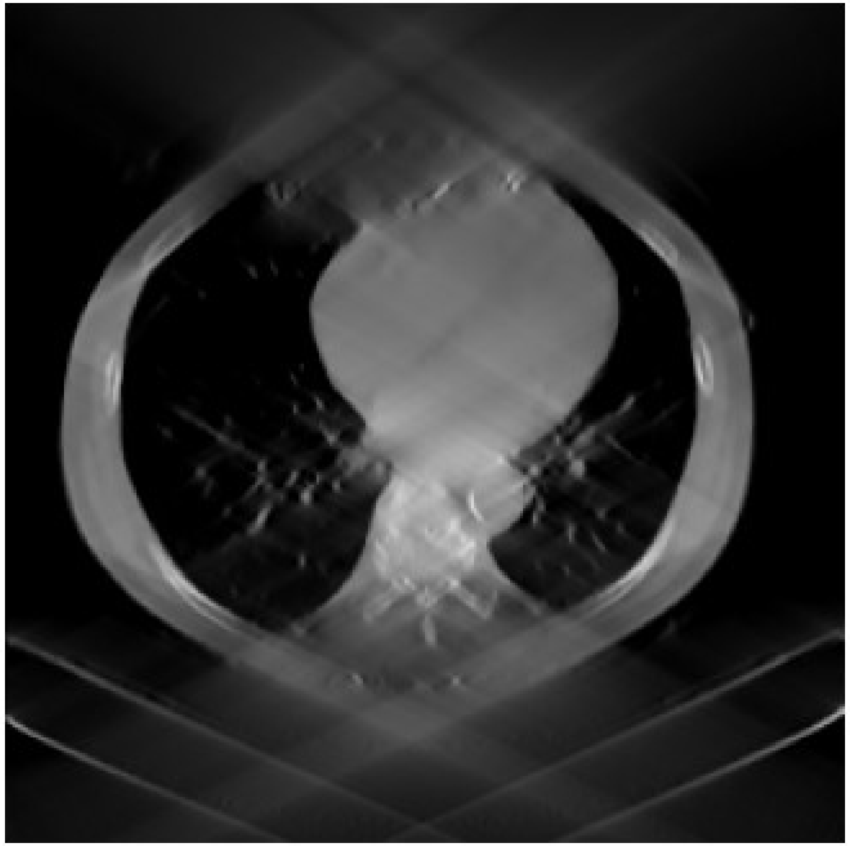
$$\varepsilon(\hat{f}) \;\leq\; \sum_{i=1}^{n} \Bigg(\prod_{j=i+1}^{n} \hat{L}_j \Bigg) \varepsilon_i .$$

Replacing $\hat{f}_k$ with the *known* f_k forces $\varepsilon_k = 0$ and strictly tightens this bound.

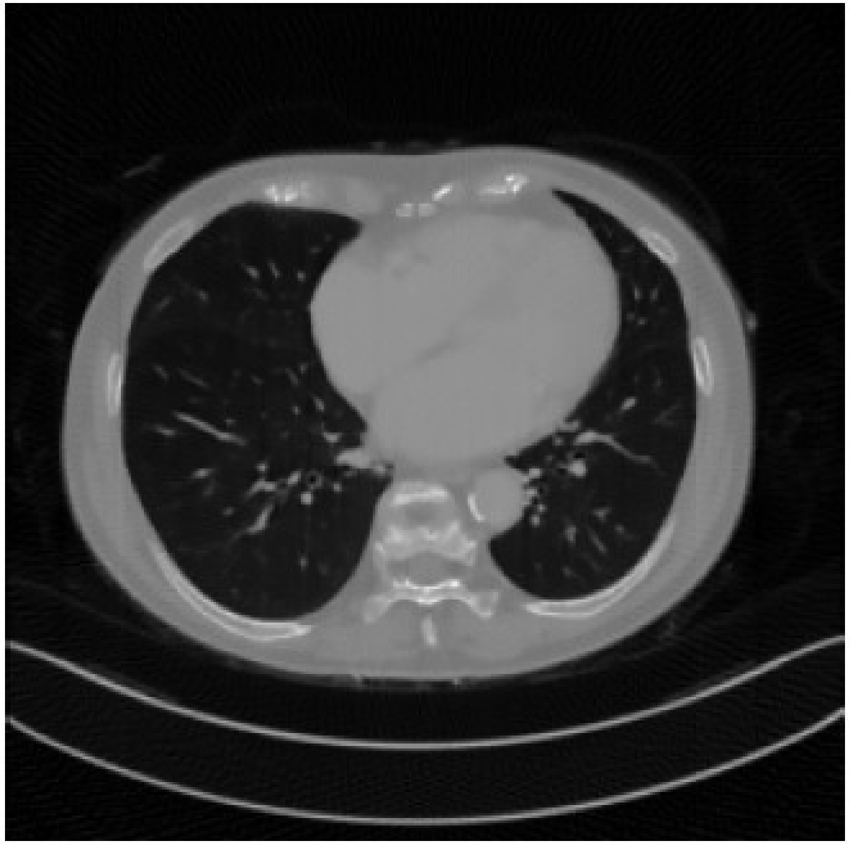
Maier et al., *Nature Machine Intelligence*, 2019.

Deep learning in medicine

Reconstruction



Conventional



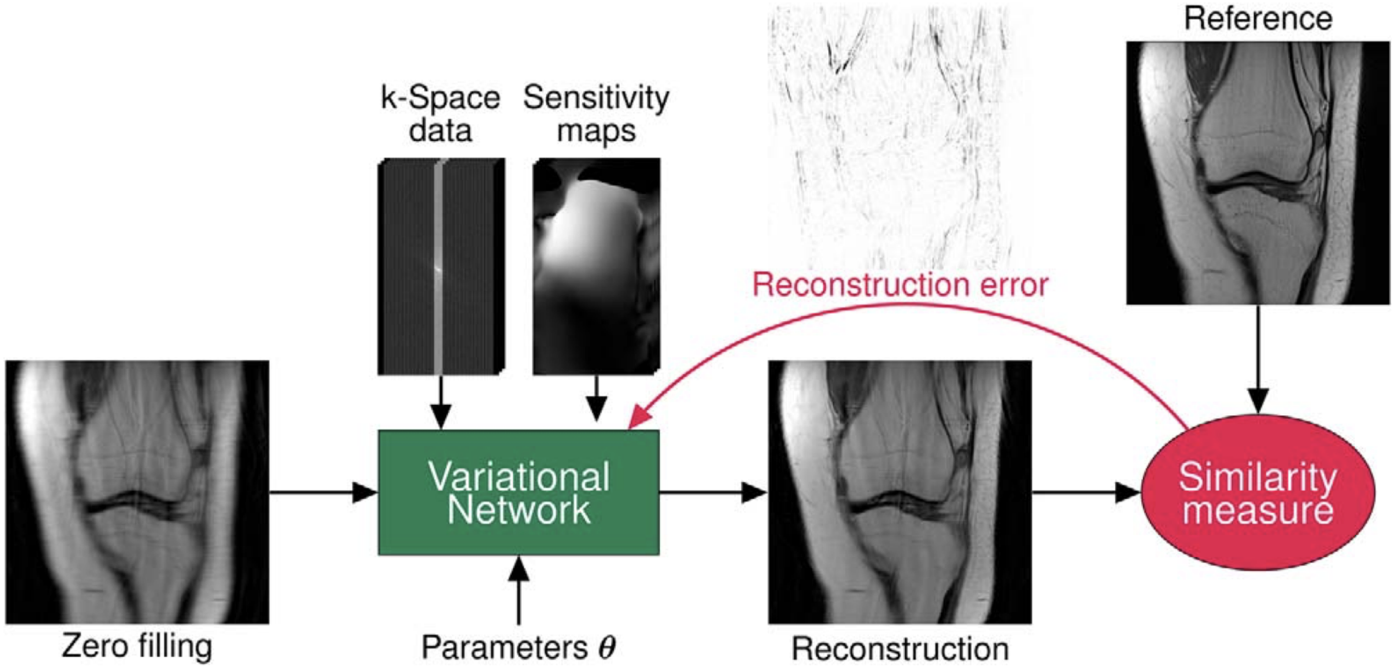
With deep learning

Deep learning in medicine

MRI variational network

Learning a Variational Network for Reconstruction of Accelerated MRI Data

Kerstin Hammernik,[1]* Teresa Klatzer,[1] Erich Kobler,[1] Michael P. Recht,[2,3] Daniel K. Sodickson,[2,3] Thomas Pock,[1,4] and Florian Knoll[2,3]



Deep learning in medicine
Recognition

Deutscher Zukunftspreis – low-field MRI made possible by deep learning.

What is agentic AI?

"How do you drive a car?"

Thought for a few seconds...

- A classic prompt to a large language model.
- After a short "thinking pause" the answer arrives as long-form running text.
- No tools, no actions – the model just *describes* how to drive.

What is agentic AI?

Long text answer of an LLM

To drive a car, follow these steps:

- **Get in and adjust:** fasten your **seat belt**, set the **seat** and **mirrors** so you can see the road clearly.
- **Start the engine:** press the **brake pedal**, turn the key or push the start button. Make sure the gear selector is in **Park (P)** or **Neutral (N)**.
- **Engage a gear:** for an automatic, move the lever to **Drive (D)**; for a manual, press the **clutch** and select **1st gear**.
- **Release the parking brake** and slowly lift your foot off the brake. Gently press the **accelerator** to move forward.
- **Steer** smoothly with both hands and **check mirrors** regularly. Keep a safe distance from the vehicle in front.
- **To stop:** ease off the accelerator, press the brake, and bring the car to a complete stop. Shift to **Park** and apply the parking brake before turning off the engine.

Always observe traffic rules and adapt your driving to road and weather.

$\Rightarrow$ **Lots of words, but no *action*.**

What is agentic AI?

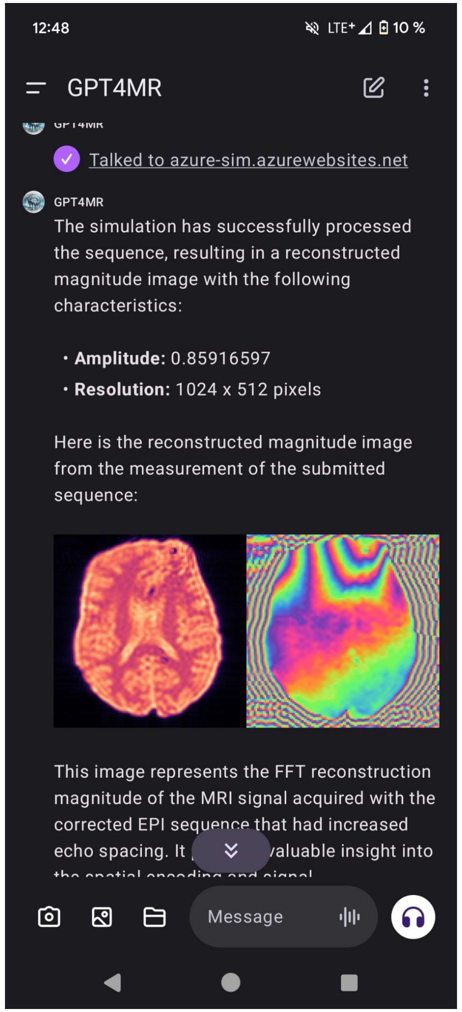
From language model to agent

Moritz Zaiss

What is agentic AI?

From language model to agent

Moritz Zaiss



Zaiss et al., *GPT4MR*, 2024.

What is agentic AI?

Task for the models

"Code a spin echo EPI (64×64, $fov = (0.2, 0.2, 0.08)\,m$, $TE = 150\,ms$)!"

- **Spin-echo EPI** – an advanced MRI sequence used for diffusion and functional imaging.
- Three models receive the *same* natural-language instruction and must produce runnable Python code.
- Contestants:

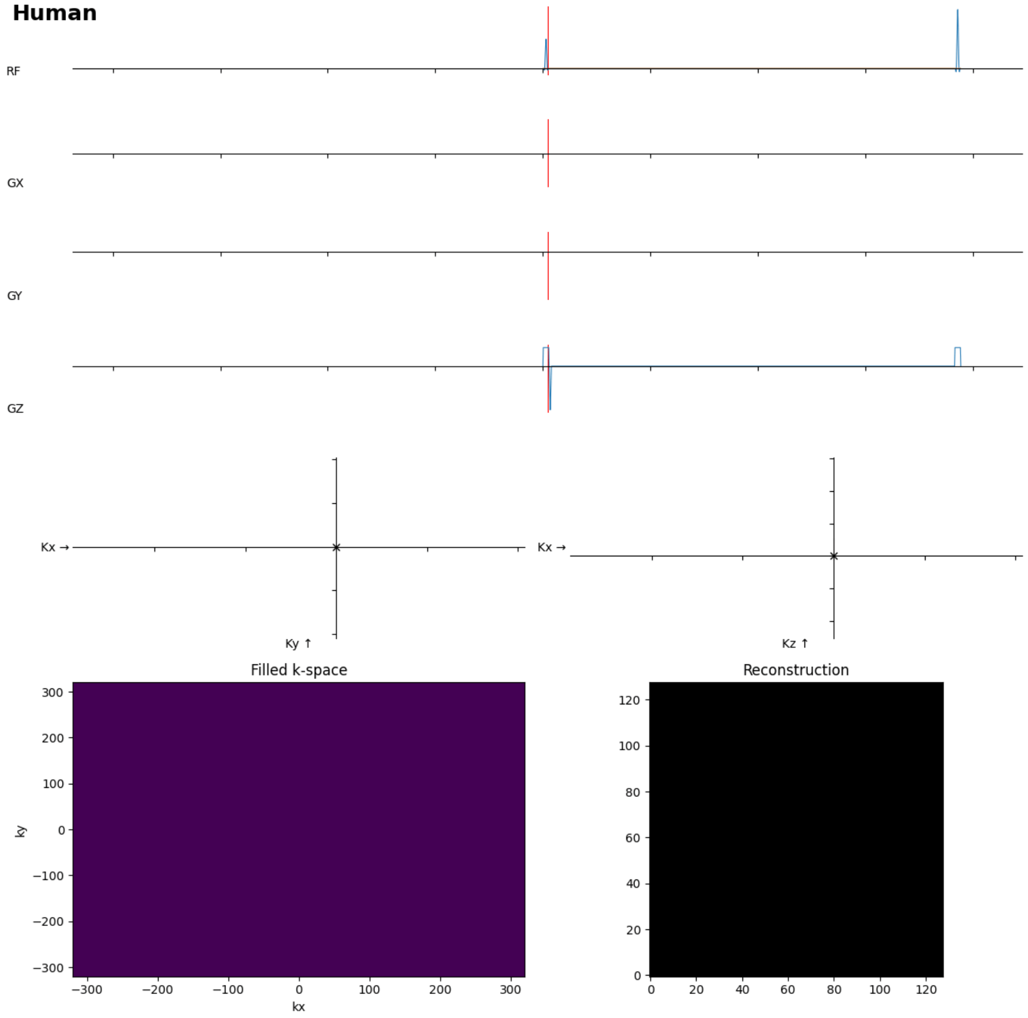
Human
reference

GPT-4o
OpenAI

GROK-3-think
xAI

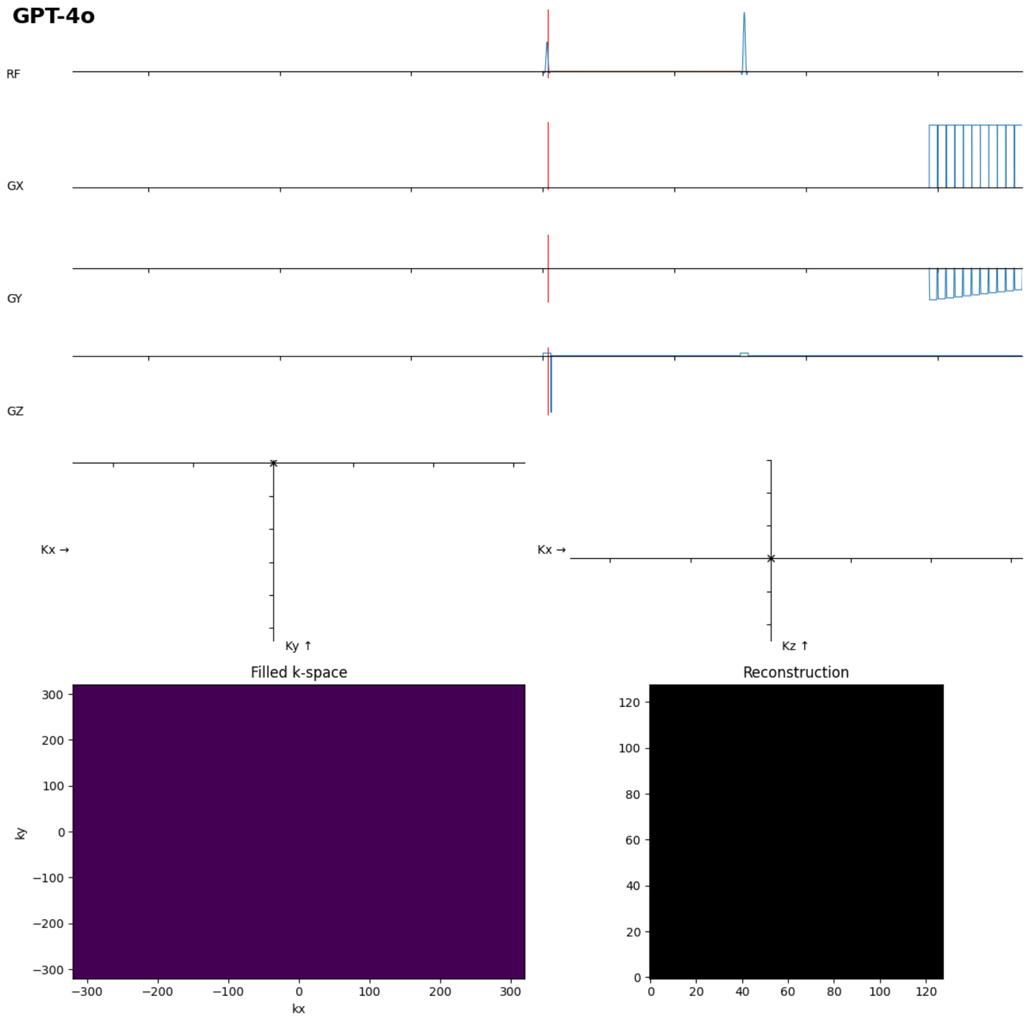
What is agentic AI?

Human baseline



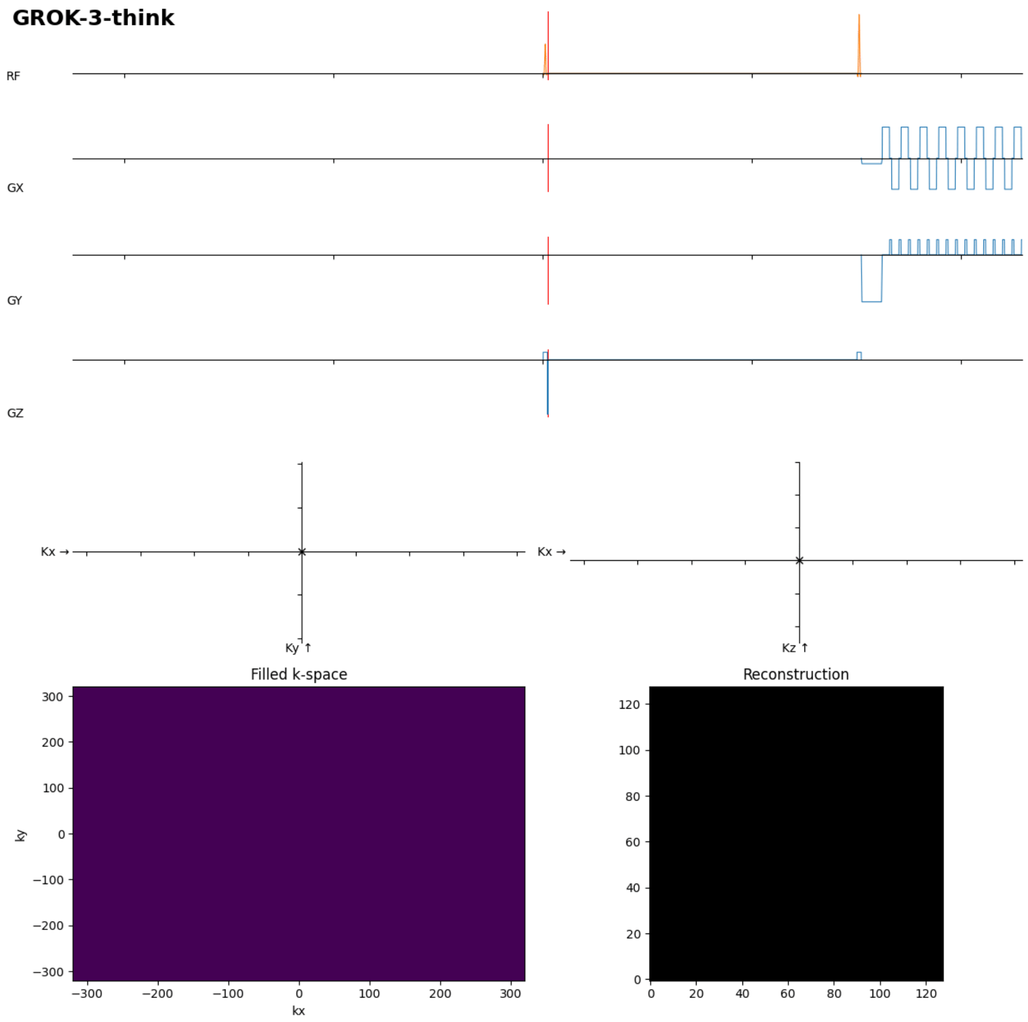
What is agentic AI?

GPT-4o attempt



What is agentic AI?

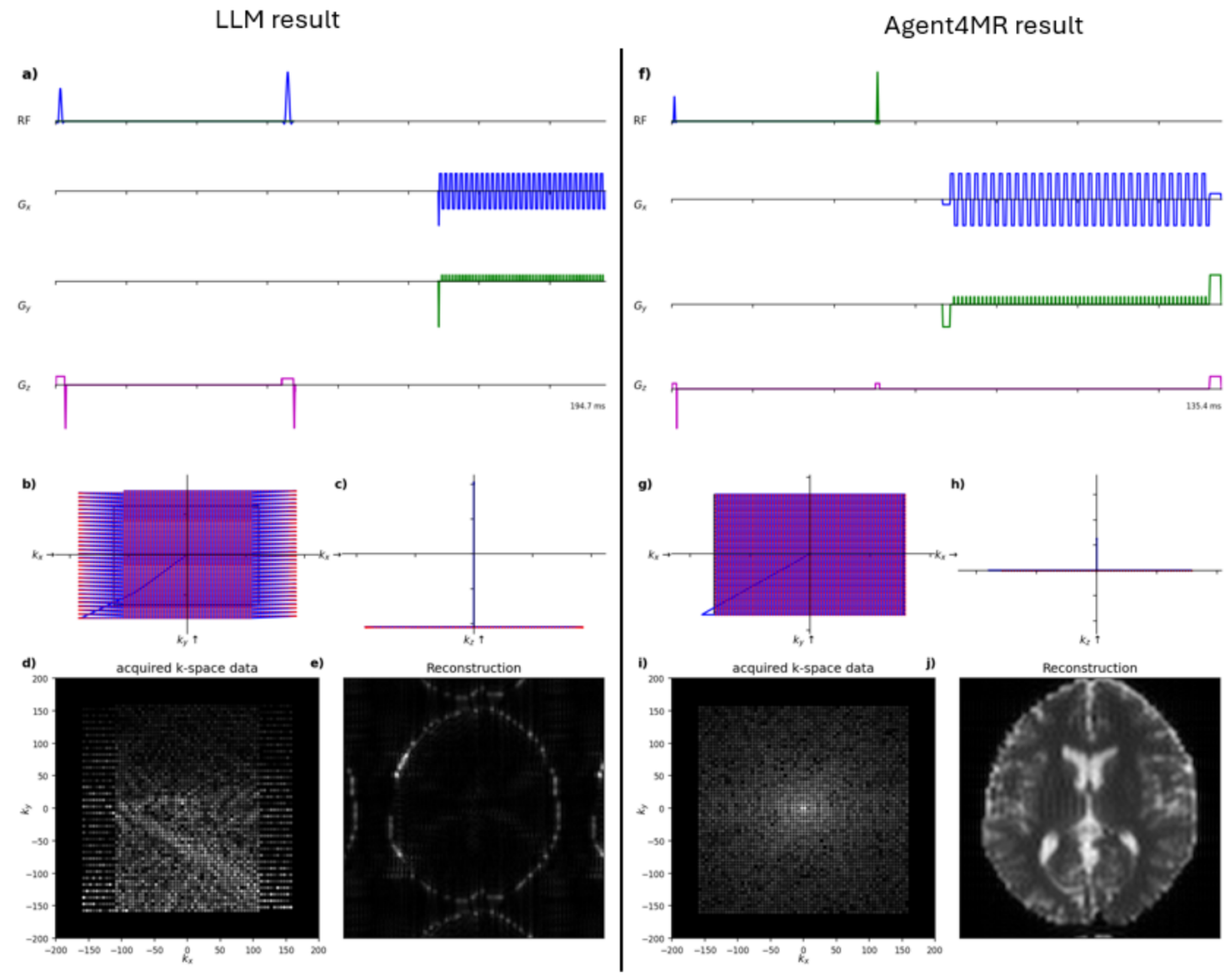
GROK-3-think attempt



How about real agentic AI?

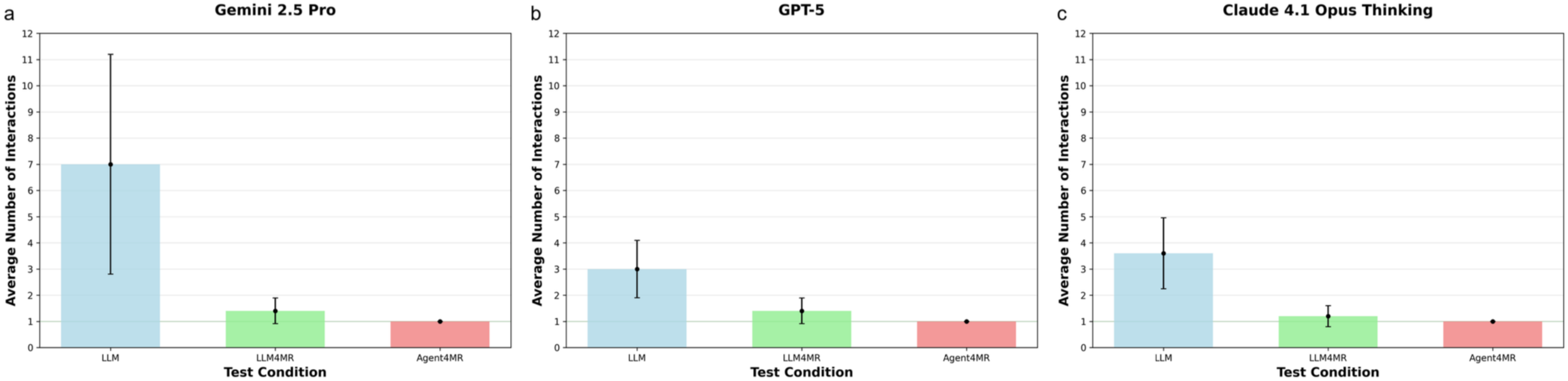
What is agentic AI?

LLM vs. Agent4MR



What is agentic AI?

Interactions per model



What is agentic AI?

Cost comparison

Table 1: User interactions N, time, token usage, and estimated cost until a satisfactory sequence. Human baseline: recently trained EPI-naive student, 2.5 h, 5 interactions.

Configuration Agent4MR	N	Time [min]	Tokens	Cost [€] [≈]
Gemini 2.5 Pro	1	4.8	87k	0.12
GPT-5	1	15	520k	0.48
Claude 4.1 Opus	1	10.2	1162k	10.00
Human	5	150	—	37.50

What is agentic AI?

Iterative progress

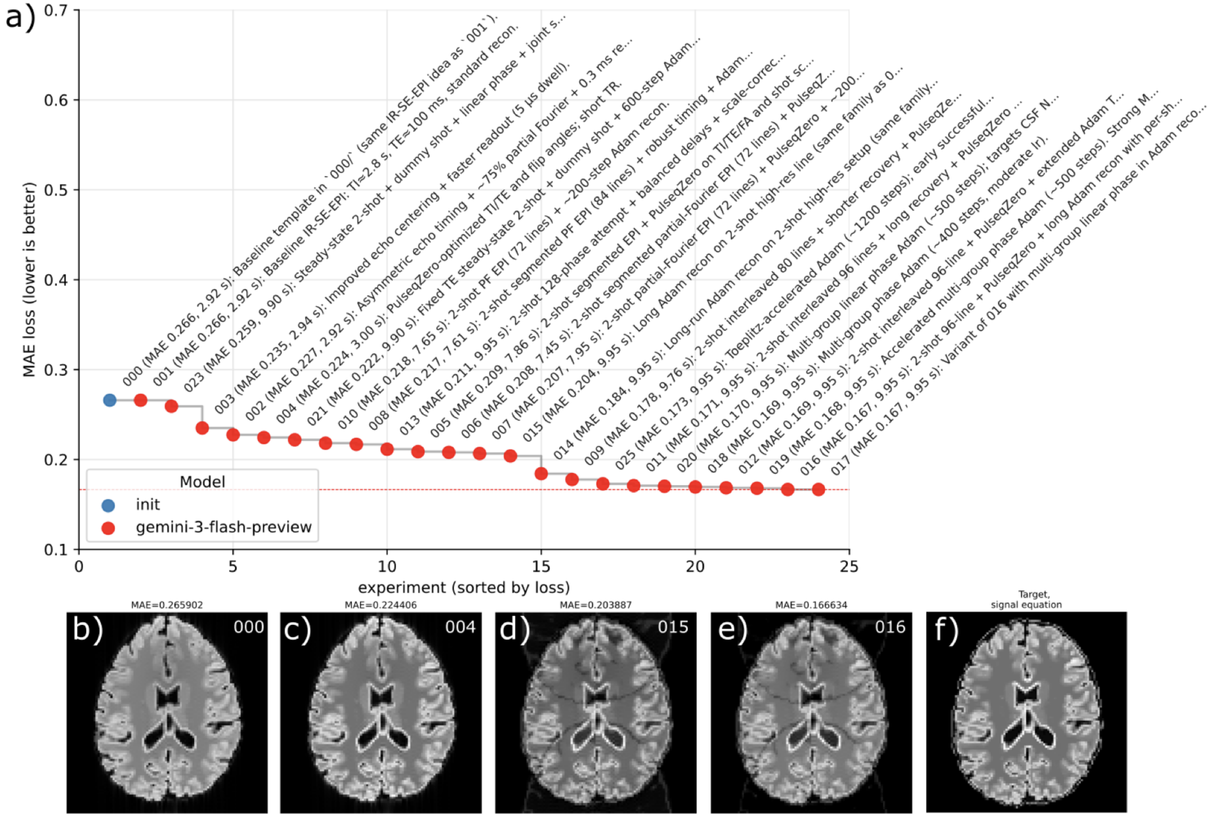


Figure 5: IR-SE-EPI MAE leaderboard progress. Autonomous agents iteratively improved the IR-SE-EPI pipeline (sequence parameters, reconstruction, post-processing). Experiments are ranked by MAE loss (worst to best). The staircase shows improvement from baseline (∼0.2659) to best (∼0.1666) through multi-window filtering, multi-shot introduction, phase correction, and differentiable TI/TE optimization.

Table 2: Normalized Token Usage and API Costs in EUR (n=25 events per model)

Gemini Model	Total Tokens [Mio]	Tokens per Exp. [Mio]	Cost [€]
2.5 Flash	25.456	1.018	2.43
3 Flash	136.340	5.453	12.85
3.1 Pro	72.039	2.881	59.72
Total	**233.836**	**9.353**	**75.00**

What is agentic AI?

Patient information

Klinikeindruck/Stempel

www.ILMTALKLINIK.DE

Patientendaten/Aufkleber

Thieme Compliance ID 15Th *GB*

Coronary Angiography, Coronary Angioplasty and Stent Implantation

(Examination of Coronary Blood Vessels, Including Possible Dilatation of Constrictions/Implantation of a Stent)

Hello,

The purpose of this informed consent leaflet is to help you prepare for the patient-doctor discussion. Please read it carefully before the discussion and complete the questionnaire carefully and completely. For better readability, we use male pronouns for occupational titles or terms referring to specific persons (e.g. doctor) but are including all genders with them.

Which procedure is planned?

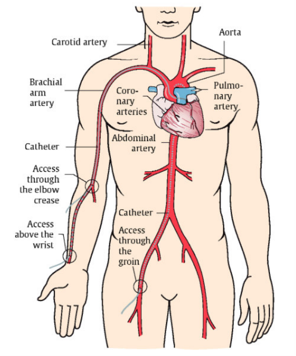


Fig. 1: Possible catheter access via the groin, above the wrist and in the crook of the arm

Current findings and your symptoms lead to suspicion of coronary artery disease.

This needs to be evaluated by a **special X-ray examination** (coronary angiography). If any constrictions (stenoses) or obstructions are found in the coronary vessels, it may be possible to perform an immediate dilatation or to reopen them using a balloon catheter (coronary angioplasty), and a stent (a tube to hold the vessel open) can be implanted if needed (stent implantation).

Are alternative examination/treatment methods available?

Other **non-invasive imaging techniques** to visualise the coronary blood vessels, e.g. computed tomography (CT) and magnetic resonance imaging (MRI), are possible but are less precise and do not enable the doctor to perform the necessary treatment measures during the same session.

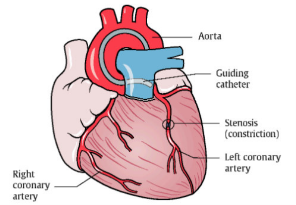


Fig. 2: Representation of coronary blood vessels showing the constriction

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What is agentic AI?

Adaptive systems

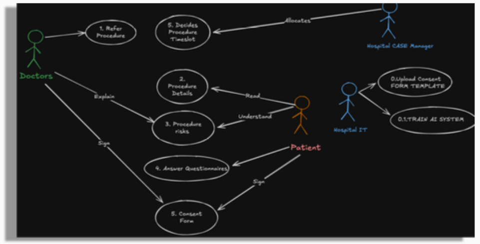
What is agentic AI?

From sketch to diagram

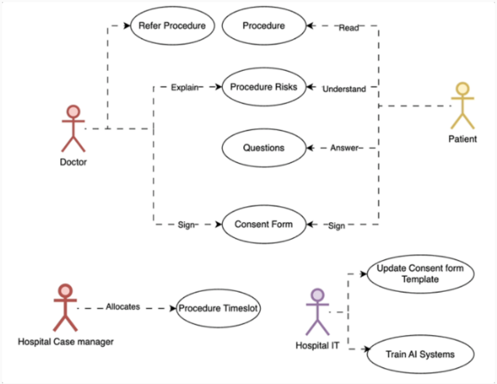
Geek Box 1: Draw.io: from sketch to formal diagram

Draw.io (also known as diagrams.net, `https://app.diagrams.net`) is a free, browser-based diagramming tool. Diagrams can be exported as XML, SVG, or PNG and version-controlled alongside code. Draw.io also supports AI-assisted diagramming: configurable large language model (LLM) backends—including OpenAI, Anthropic, and local models—can generate, modify, and explain diagrams from natural-language prompts (`https://www.drawio.com/doc/faq/configure-ai-options`).

The two figures below show the same actor model: first sketched by hand on an iPad, then redrawn in Draw.io. The content is identical—the difference is the tool.



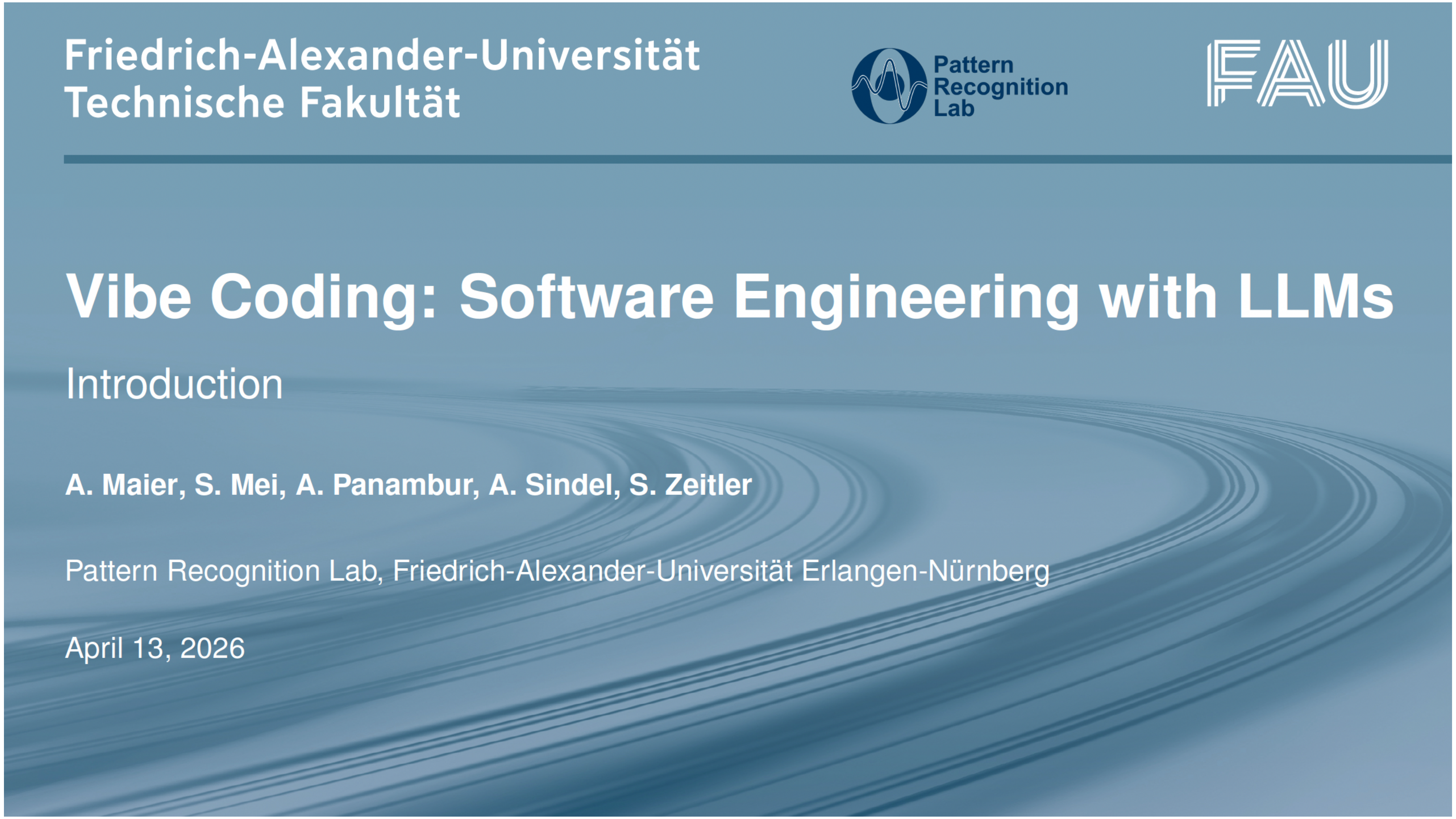
(a) iPad sketch: unstructured brainstorming, needs vision-language model (VLM) for access.



(b) Draw.io: version-controllable, exportable, and AI-editable.

What is agentic AI?

Vibe coding



What is agentic AI?

Advanced AI-assisted techniques

Advanced AI-Assisted Techniques

virtuelle hochschule bayern Pattern Recognition Lab FAU

Claude Skills [3]:

- Shareable workflow automation (YAML + Markdown)
- Team-wide best practices as code
- Deployment, testing, code review
- Available at `github.com/anthropics/skills`

Model Context Protocol (MCP):

- Standardized tool integration
- Connect AI to databases, APIs, files
- Growing ecosystem of plugins

Other Advanced Patterns:

- **Agentic workflows**: Multi-step autonomous tasks
- **Tool-augmented coding**: AI uses external tools
- **Vision support**: Screenshot-driven debugging

Key Benefit

These techniques enable AI to perform complex, multi-step workflows beyond simple code generation.

to a programmer interface for an agent.

What is agentic AI?

Markdown-driven quality gates

Geek Box 13.3: Markdown-driven quality gates: when agents run the production checklist

A key insight from agentic software engineering is that quality criteria, which used to require human inspection, can now be expressed as structured markdown instructions that an AI agent executes autonomously. The agent reads the checklist, runs the specified checks, collects all findings, and presents them for human review—combining the thoroughness of automated scanning with the judgment-level assessments previously exclusive to human reviewers.

Example: a production checklist for book chapters. This book uses a markdown-based production checklist that an AI coding agent runs before each chapter is finalized. A condensed excerpt:

```
# Production Checklist for Chapter Review
The agent should: 1. Detect all problems first.
2. Present a PASS/FAIL table. 3. Ask the user
which to fix. 4. Apply only selected fixes.

## Checks to Run
### Cross-References
- All figures/tables/geek boxes referenced in text.
### Accessibility
- All figures have descriptive alt-text.
### Layout
- No "Float too large" or overfull hbox warnings.
### Acronyms
- No duplicate acronym expansions.
```

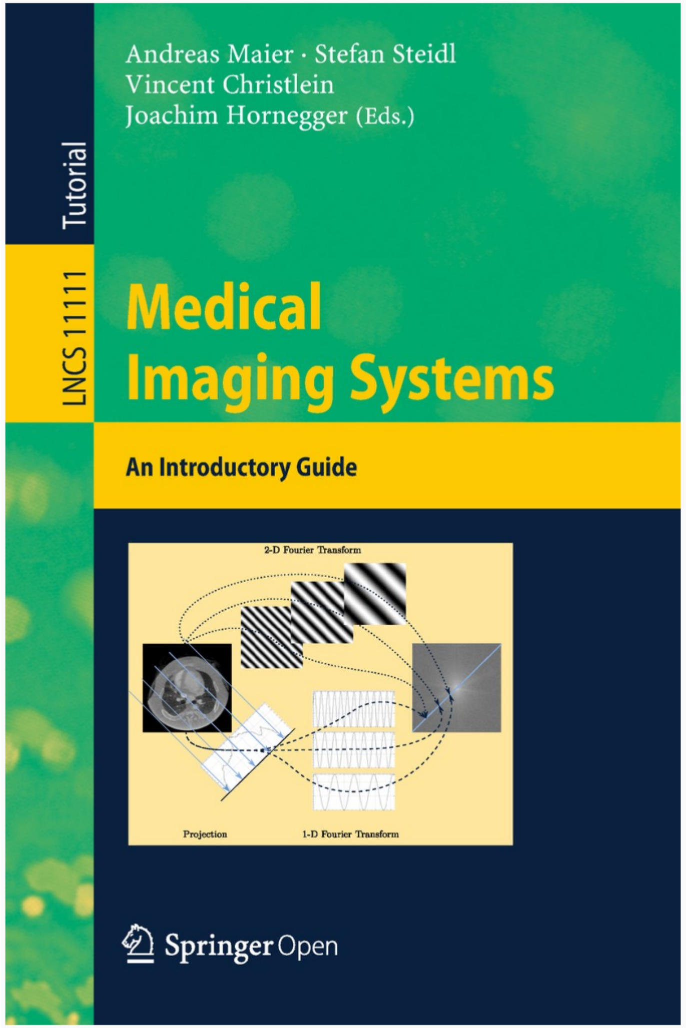
Several of these checks—alt-text quality, duplicate acronym detection, caption descriptiveness—require understanding the *meaning* of text, not just its syntax. An AI agent can evaluate whether alt-text actually describes what a figure shows or whether a manually spelled-out acronym conflicts with a `\gls{}` expansion nearby. These are assessments no simple script can perform reliably.

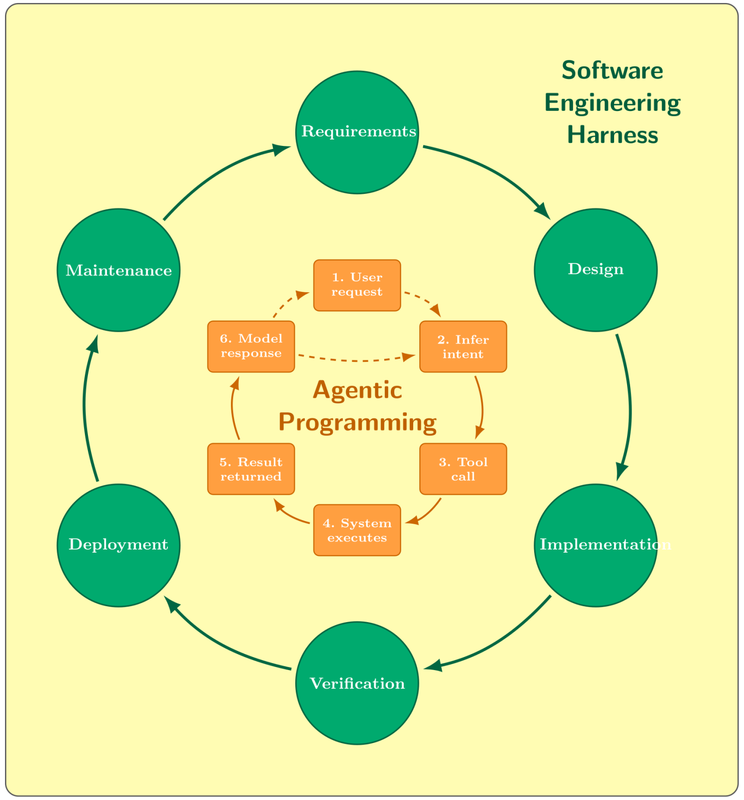
The detect-then-ask workflow. The checklist specifies a four-step process: detect all issues, present a PASS/FAIL summary, ask the user which fixes to apply, and then execute only approved changes. This separation of detection from correction mirrors the role-separation principle from Chapter 12: the agent that finds problems should not unilaterally fix them. The human decides which findings are genuine issues and which are acceptable.

This pattern generalizes beyond book production. Any domain with explicit quality criteria—code review checklists, regulatory compliance, accessibility audits, documentation reviews—can be encoded as markdown and executed by an agent. The quality gate becomes a reusable, version-controlled artifact rather than tacit knowledge in a reviewer's head.

What is agentic AI?

Textbook





What is coming next?

Almost everything is easy now. . .

China and AI

Almost everything is easy now…

Conference shares

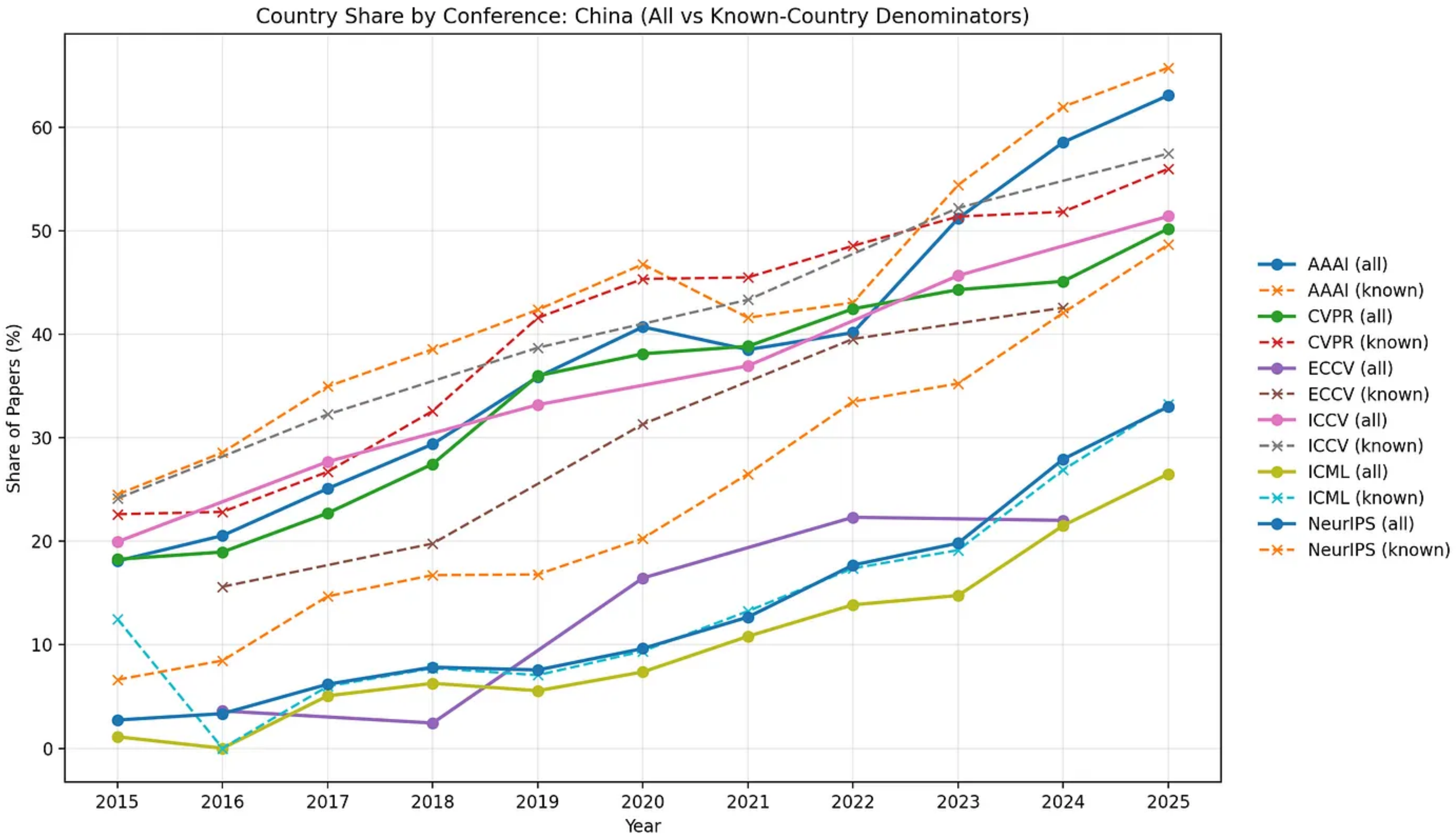
Conference	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
CVPR	~30%	32%	35%	35%	39%	39.2%	~40%	~38%	~39%	~40%
ICCV/ECCV	~30%	30%	32%	35%	~37%	~40%	~40%	~38%	~40%	~40%
ICML	~10%	12%	15%	18%	~20%	~22%	~25%	~28%	~30%	~30%
NeurIPS	8%	10%	12%	15%	*~15%*	~18%	~20%	~25%	~28%	~30%
AAAI	15%	18%	22%	28%	~35%	50%	~50%	~45%	~48%	~50%

Almost everything is easy now. . .

ConferenceStats workflow

Almost everything is easy now. . .

Conference contributions from China



Agent4CT

All deep-learning CT methods, in one weekend

- Continuously-running LLM agent that edits a CT reconstruction codebase, runs short GPU experiments, and keeps changes only when the metrics agree. Five agents run in parallel across five CT challenges and pool findings through a shared scratch pad.

Re-implemented solvers (`pentathlon/demo_dl_reference`):

- **Analytical:** PyroNN FBP (Syben 2019).
- **Classical iterative:** TV (iterative, supervised, hyper-search); Wu *et al.* 2015 (+ trainable port).
- **Variational / unrolled:** Hammernik 2017 variational network (+ CT port); Learned Primal-Dual (Adler & Öktem); ITNet v1–v3.
- **Self-supervised denoising:** dual-domain U-Net and trainable bilateral filters (Wagner 2023 / 2022, with supervised variants).
- **Per-scene neural fields:** NAF (neural attenuation field); R2Gaussian.
- **Foundation models:** RAM (Terris 2025).
- **Diffusion priors:** DDPM; diffusion-prior sampling; diffusion-recon with data consistency.
- **Transformer:** U-Swin.

`github.com/akmaier/Agent4CT`

Agent4CT

From paper to benchmark – a structured loop

- **Tight iteration budget.** 5 min per GPU experiment forces the agent to trade architecture against throughput.
- **Stage checks every 30 iterations.** 1 h validation on a $3\times$ larger subset catches overfit before it propagates.
- **Git-backed journal.** Every kept change is a commit; discards are `git reset` plus a row in the experiment TSV.
- **Shared scratch pad.** `observations.jsonl` pools rules-of-thumb across the five agents.
- **Anti-overfit rules** bootstrapped into every agent's `program.md` – so each new run starts on the others' shoulders, not from scratch.

Agent4CT

The Pentathlon – five CT leaderboards

- AAPM 2016 Low-Dose CT (Mayo)
- AAPM 2021 DL Sparse-View CT (128-view fan-beam)
- AAPM 2022 TrueCT
- AAPM 2022 DL-Spectral CT
- AAPM 2024 CT Metal Artifact Reduction

Score: *headroom recovered* $\in [0, 1]$; 1 matches the oracle.

Current best – Breast CT

Learned Primal-Dual
SSIM **0.9062**
at 1.49 M parameters

References

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- Conference statistics tooling: https://github.com/akmaier/ConferenceStats
- Agent4CT – agentic re-implementation of deep-learning CT methods: https://github.com/akmaier/Agent4CT

Thank you!

Slides on GitHub

github.com/akmaier/AIMed2026Slides/tree/dldxhealth-2026